

Sentiment Classification & Summarization

E-Commerce Product Reviews Using NLP

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Problem Statement

- E-commerce platforms generate massive volumes of customer reviews
- Manual analysis is time-consuming and impractical
- Star ratings alone cannot capture nuanced opinions
- Need automated solution for sentiment classification and summarization

Project Objectives

- Develop end-to-end NLP pipeline for review analysis
- Perform sentiment classification (positive/negative)
- Generate concise one-sentence summaries
- Integrate both tasks for practical e-commerce use

Technical Stack

Sentiment Analysis

Fine-tuned BERT model for contextual understanding and high-accuracy binary classification

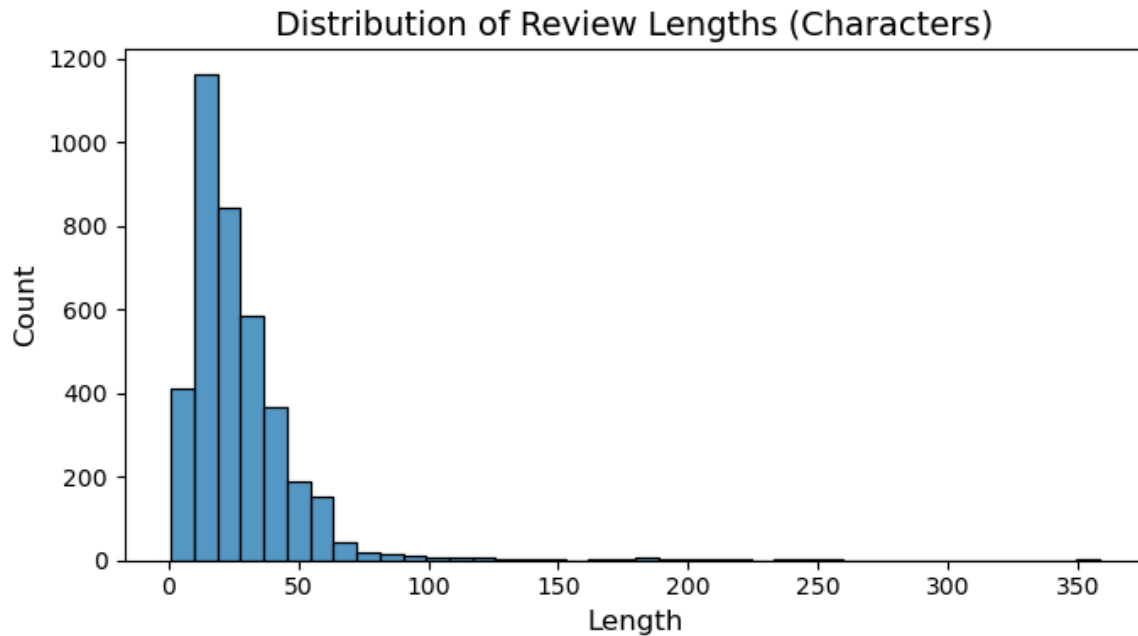
Abstractive Summarization

Pre-trained BART model for generating fluent, concise summaries

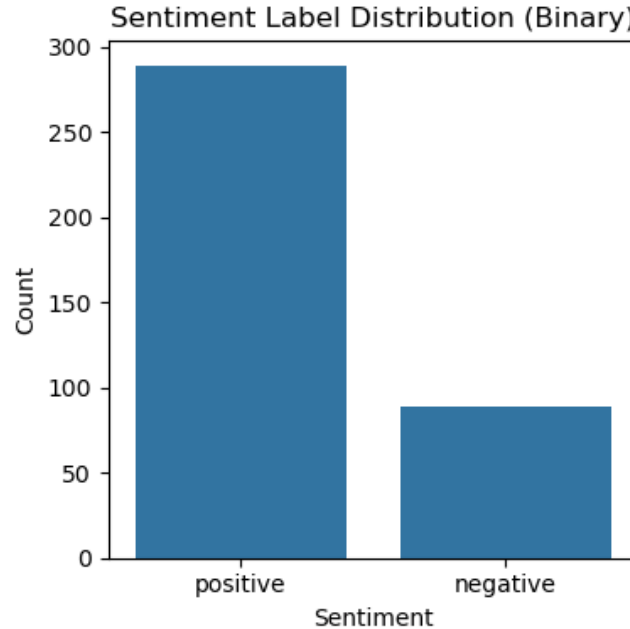
Data Preparation

- **Training Dataset:** Kaggle e-commerce reviews (3,841 cleaned reviews)
- **Testing Dataset:** NLTK product reviews (unlabeled, 99 reviews per product)
- **Preprocessing:** Lowercase, remove punctuation, filter empty reviews
- **Mapping:** 1-2 stars → negative, 4-5 stars → positive

Dataset: Review Length Distribution



Dataset: Sentiment Distribution



Baseline Models Comparison

TF-IDF + Logistic Regression

Accuracy: 79%

F1-score: 0.88

Features: 30K max features

TF-IDF + Linear SVM

Accuracy: 88%

F1-score: 0.93

Best baseline

Both use TF-IDF vectorizer with unigrams and bigrams

BERT Fine-Tuning Architecture

- **Model:** bert-base-uncased (12 layers, 110M parameters) + 2-class head
- **Training Hyperparameters:** 3 epochs, lr=2e-5, batch_size=16, optimizer=AdamW
- **Input Processing:** Max length 128 tokens, padded/truncated as needed
- **Performance:** 81.6% accuracy on validation, F1-score 0.89 (limited by 2K training samples)

BART Abstractive Summarization

- **Model:** facebook/bart-large-cnn (406M params, seq2seq architecture, CNN pre-trained)
- **Decoding Strategy:** Beam search (num_beams=4), max_length=40, early_stopping=True
- **Preprocessing:** Input truncated to 512 tokens, padding applied
- **Evaluation:** ROUGE-1: 0.6934, ROUGE-2: 0.6549, ROUGE-L: 0.6841 (heuristic refs)

Final Results & Evaluation

- **BERT on NLTK Test:** 91% accuracy, F1-score 0.90 (22 misclassifications in 76 samples)
- **Confidence Distribution:** 85%+ for majority, lower for ambiguous mixed-sentiment reviews
- **BART Summary Quality:** Fluent, concise, captures key aspects, some detail loss in complex reviews
- **End-to-End Pipeline:** Successfully integrates both models with < 1s inference per review

Pipeline Example

Original: "I love the sound quality but uncomfortable after long use"

Sentiment: Positive (0.89 confidence)

Summary: "Great sound but uncomfortable for long use"

Error Analysis & Root Causes

- **Mixed Sentiments:** Reviews like "great product but bad customer service" → lower confidence (0.55-0.70)
- **Summarization Detail Loss:** Long reviews (>200 chars) lose nuance; BART truncates to capture main idea only
- **Short Review Ambiguity:** Reviews < 20 chars (e.g., "okay", "decent") → 50-60% confidence due to limited context
- **Dataset Imbalance:** 3.25:1 positive-to-negative ratio causes systematic bias; multi-class approach recommended

Future Directions & Scalability

- **Multi-Class Sentiment:** Add neutral class; use weighted loss to handle imbalance (positive/neutral/negative)
- **Domain Adaptation:** Fine-tune on category-specific reviews (electronics, clothing, food) for better generalization
- **Enhanced Evaluation:** ROUGE metrics for summaries; human evaluation studies; aspect-based sentiment analysis
- **Production Deployment:** REST API (FastAPI), batch processing with Airflow, monitoring with MLflow

Practical Applications

For Shoppers

Quick understanding of product feedback and sentiment trends

For Sellers

Monitor and analyze large volumes of reviews efficiently

Conclusion

Successfully developed a robust NLP pipeline combining BERT and BART

Achieved 91% accuracy in sentiment classification

Generated fluent, concise summaries capturing review essence

Demonstrated practical value for e-commerce analytics